



**USERS MANUAL
FOR
SSQ-53F/GPS SONOBUOY**

P/N 90024A



Procuring Activity: NSWC – Crane, IN

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Document Changes

- | | |
|---------------|--|
| 2.3, 3/9/06 | Corrected error in Subcarrier frequency and logic level convention in Table 1. |
| 2.4, 12/13/06 | Added diagnostic mode command to turn GPS subcarrier On and Off. |
| 2.5, 2/4/08 | <ul style="list-style-type: none">- Corrected error and cleaned up references to subcarrier frequency language (noted early in document as 35 KHz, corrected to 33.5 KHz)- Defined uplink data idle state |
| 2.6, 8/6/09 | Corrected GPS Subcarrier mark and space frequencies |
| 2.7, 4/14/10 | Corrected error in Table 2 Sonobuoy Channels that interfere with GPS Operation. |
| 2.8, 4/25/11 | Added note describing NMEA GPS Message time tag UTC time issue |

1 Overview

This document provides reference information for the special SSQ-53F sonobuoys that have been manufactured by Ultra Electronics – Undersea Sensor Systems Inc. The sonobuoy meets all the requirements of the Production Sonobuoy Specification (PSS) for the AN/SSQ-53F and adds GPS and Extended Calibrated Omni functionality. The SSQ-53F/GPS Sonobuoy is a modified version of the USSI FY04 SSQ-53F Sonobuoy. The baseline sonobuoy has been modified to,

- 1) Add a GPS antenna and receiver
- 2) Update the upper unit DSP to generate the GPS Sub-Carrier and provide selectable CO bandwidth.
- 3) Integrate the GPS hardware with the current upper electronics

The sonobuoy transmits analog acoustic and GPS data via a VHF analog FM uplink. The acoustic data channel is selectable via Electronic Function Select (EFS) or Command Function Select (CFS). The sonobuoys are provided for Over-The-Side use only.

2 Sonobuoy Operating Modes

The sonobuoy is configured to operate in DIFAR, Calibrated Omni-directional (standard or extended CO), or Constant Shallow Omni-directional (CSO) Acoustic Uplink with simultaneous GPS data subcarrier. During standard CO operation the GPS subcarrier is centered at 33.5 KHz. The DIFAR, CSO, and CO uplink signals with GPS are shown in Figures 1 through 3.

An Extended Calibrated Omni (XCO) capability has been added to all FY-04 or later versions of the Ultra Electronics – USSI Q53F/GPS sonobuoy. This mode is selectable via EFS prior to deployment or CFS post deployment. The XCO mode supports the uplink of calibrated acoustic signals from the CO hydrophone up to 40 KHz (see Figure 4 and 5). To support the larger acoustic bandwidth in XCO mode, the GPS subcarrier is centered at 45 KHz. The subcarrier is centered at 33.5 KHz for all standard acoustic modes (DIFAR, CSO, and CO).

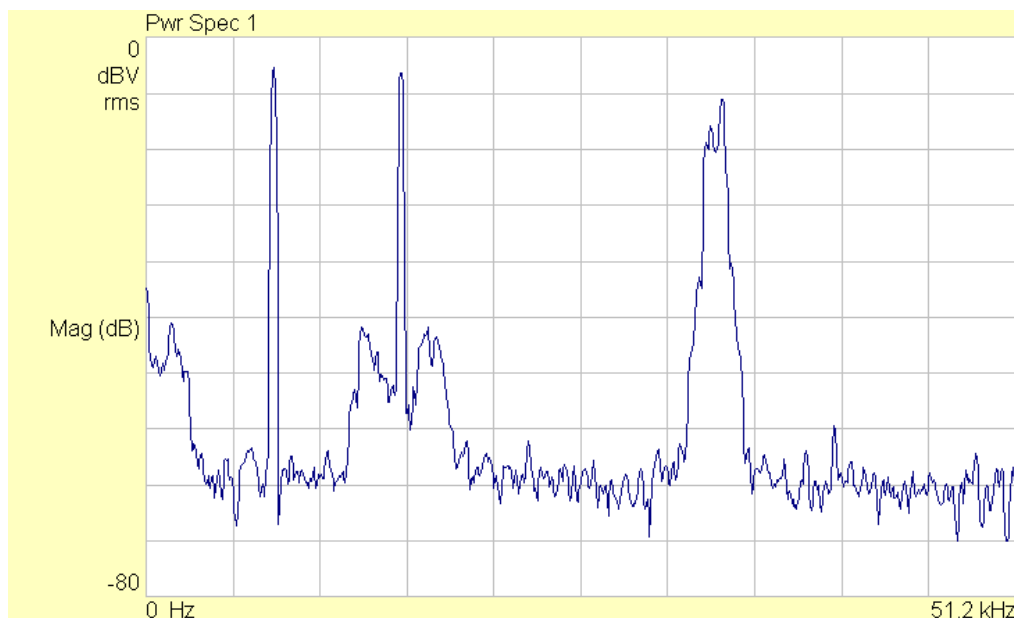


Figure 1. DIFAR Acoustic and GPS Subcarrier

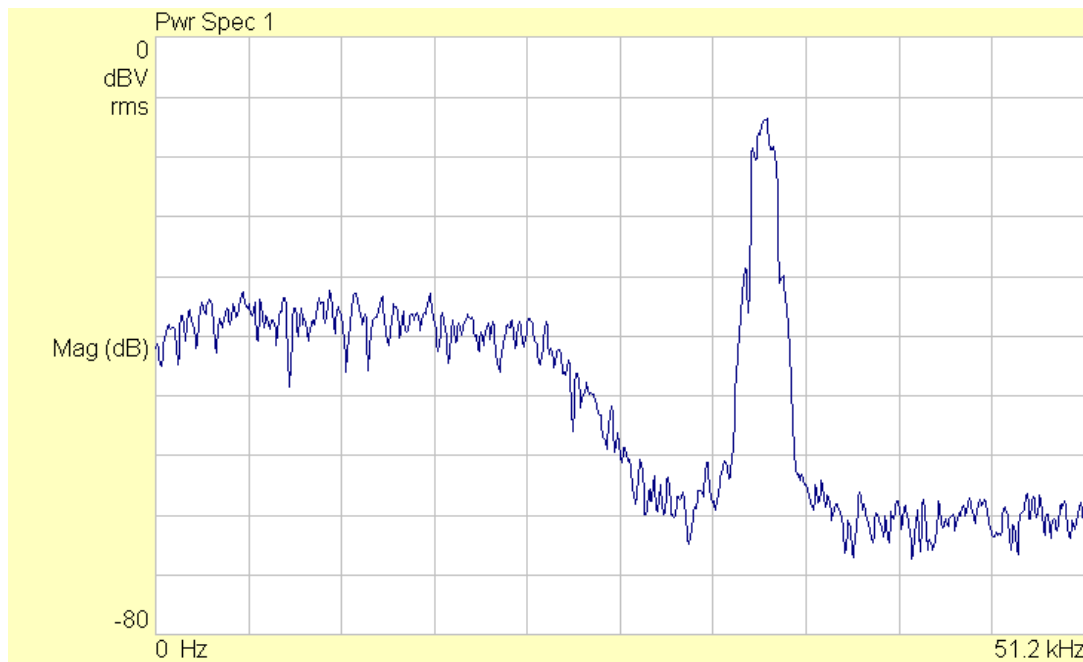


Figure 2. Standard Calibrated Omni and GPS Subcarrier

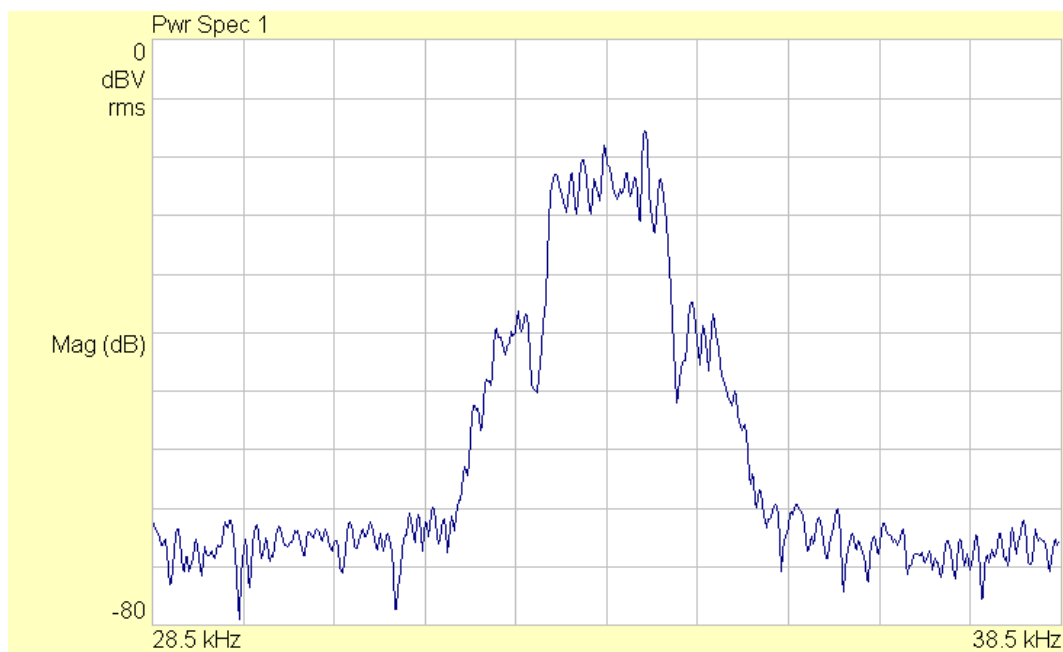


Figure 3. GPS Subcarrier (DIFAR, CSO, and CO modes)

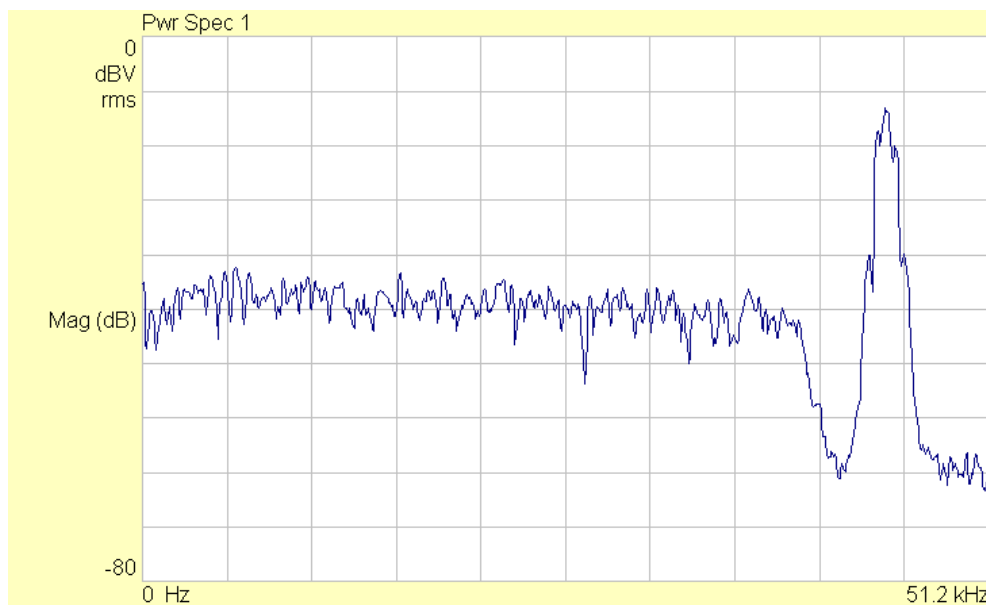


Figure 4. Extended Calibrated Omni and GPS Subcarrier

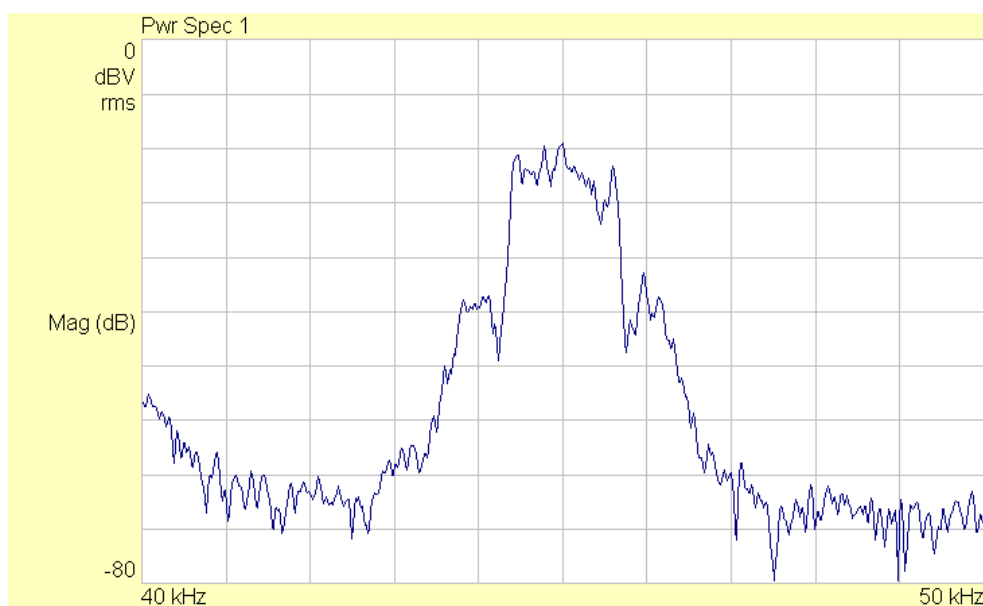


Figure 5. FSK Subcarrier (Extended CO mode)

The buoys will start in the acoustic mode and RF Channel selected by the EFS. A description on how to select parameters via EFS is shown in Appendix A. The DIFAR, CSO, and standard CO buoy acoustic processing modes are identical to the standard SSQ-53F sonobuoy. If the buoy is recovered and re-used the RF Channel will update to the RF channel it was operating on when it was recovered (ie if the buoy RF channel was changed via CFS). The acoustic mode will always revert to the mode selected by EFS prior to deployment.

3 Hydrophone Calibration

3.1 Standard SSQ-53F CO Calibration

The standard buoy is specified to have an acoustic sensitivity of 19 KHz deviation for 116 dB/1uPa Sound Pressure Level (SPL) input at 100 Hz. At other frequencies the buoy sensitivity is defined by the acoustic response and calibration tables provided with each delivery. The acoustic response is defined in the Production Sonobuoy Specification (PSS).

3.2 Extended SSQ-53F CO Mode

The extended CO mode available in the SSQ-53F/GPS Sonobuoy is selected via EFS (a manufacturer only CFS command found in Appendix C). The acoustic transfer functions for standard and extended CO modes are shown in Figure 6.

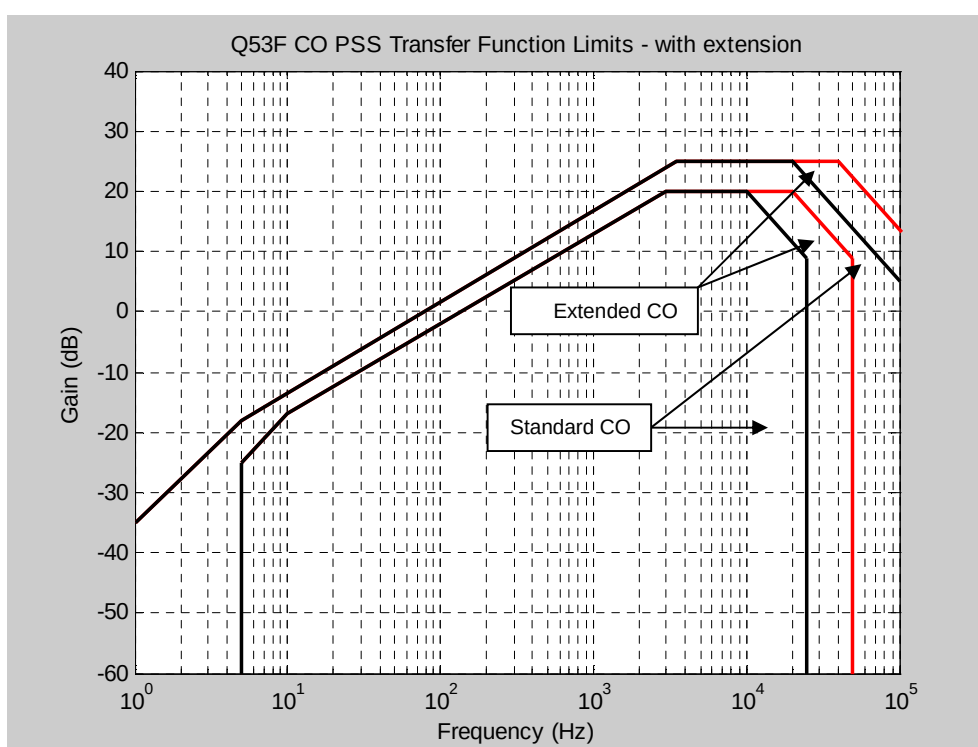


Figure 6. SSQ-53F/GPS Standard and Extended CO Transfer Functions

4 GPS Data

4.1 Data Modulation - Subcarrier

The GPS data Frequency Shift Keying (FSK) modulates a subcarrier that is summed with the acoustic data prior to modulation of the VHF carrier. The GPS Uplink data settings are 1200 baud, 8 bits, no parity, and one stop bit. As in other UART implementations, the data word is sent LSB first.

The FSK subcarrier frequency is fixed for DIFAR and CSO acoustic modes. The subcarrier frequencies are automatically configured per the selected CO mode (Standard or Extended). The modulation index for the FSK modulation in both modes shall be 0.7. The subcarrier frequencies for all the modes are shown in Table 1.

Acoustic Mode	Subcarrier Center Frequency	Space (Logic "0")	Mark (Logic "1")
DIFAR, CSO, and CO ¹	33.5 KHz	33.08 +/- 0.05 KHz	33.92 +/- 0.05 KHz
XCO ¹	45.0 KHz	44.58 +/- 0.05 KHz	45.42 +/- 0.05 KHz

Notes:

- 1) CO mode is standard 20 KHz bandwidth and XCO is extended 40 KHz bandwidth
- 2) The modulation index for the FSK modulation is 0.7.

Table 1. FSK Subcarrier Frequencies

The idle state of the subcarrier when no data is being sent shall be Logic "1".

The frequency deviation of the VHF carrier by the GPS subcarrier shall be 7.5 +2.5 -3.1 KHz (+/- 2.5 dB).

The subcarrier modulation shall not generate signals greater than -40 dB rel to 7.5 KHz deviation at +/- 1 KHz of the carrier frequency and -60 dB rel to 7.5 KHz deviation at +/- 2 KHz of the carrier frequency.

The subcarrier may be turned off via CFS using the GPS Subcarrier ON/OFF command found in Appendix C.

4.2 Data Messages

The standard GPS data transmission uses the standard NMEA-183 ASCII messages GSA, GSV, and GGA (an optional replacement of the GGA word with RMC is available). The units can be factory configured to uplink other NEMA-183 words upon request. Note, the configuration of all units manufactured from 1/1/06 are configured to uplink messages GSA, GSV, GGA, and RMC. This change is being made to standardize the GPS buoy configurations used by various groups of the US Navy. The GPS messages are defined in Appendix B. The messages are repeated every 5 seconds. Note, the GSV message transmitted every 5 seconds may contain multiple data lines if data for more than 4 satellites is provided.

4.2.1 GPS NMEA Message Time Tags

This section has been included to clarify the time tag output for NMEA messages used with the GPS buoy. All messages output time in UTC. However, at the beginning of each buoy deployment, the GPS receiver may output messages in GPS time prior to reception of the navigation message (ie cold start). The GPS satellite transmissions received by the GPS receiver are referenced to "GPS time". GPS time is a system time reference and differs from UTC time. The difference is caused by the number of leap seconds that have been added to UTC time since the start of the GPS time reference (midnight of January 6, 1980). Currently the time offset for GPS to UTC time is 15 seconds. The UTC time offset is sent to the user as part of the navigation message. This navigation message is 1500 bits long and is transmitted at a rate of 50 bits per second. Depending on when the GPS receiver is powered on, it can

take up to 12 ½ minutes to receive the message. The GPS receiver time to first fix is approximately 30 seconds and it will generate NMEA messages with a time tag. If the UTC time offset message has not been received the time tag output is in GPS time. The sonobuoy does not monitor for this condition and sends the NMEA messages straight from the GPS receiver to the user. Also, the status of the time tag output is not indicated by the status values in the NMEA messages. Upon receiving a valid UTC time message, the next GPS time tag will be referenced to UTC time and the user will see a jump (15 sec) in the time tag. This condition will occur only once during the life of the sonobuoy.

4.3 Buoy VHF Uplink

The buoy VHF uplink will interfere with GPS operation on the channels listed below. Also, per the PSS specification channels 93, 57, and 58 are not available for selection via EFS or CFS.

Transmit Channels			
49	52	87	90
50	53	88	91
51	54	89	92

Table 2. Sonobuoy Channels that interfere with GPS Operation

4.4 Buoy UHF Downlink (CFS Commands)

The sonobuoy shall responds to the CFS commands specified in PSS Appendix E for the SSQ-53F (duplicated in Appendix C of this document).

5 GPS SONOBUOY HANDLING PROCEDURES

5.1 Electronic Function Select (EFS)

EFS SET and VERIFY switches are access from the front of the unit. The procedure for setting the EFS is found in Appendix B.

5.2 Buoy Deployment and Recovery

The sonobuoy can be deployed by hand over the side or air launched. Special care should be taken when hand launching units to insure that the float and signal cable are not damaged by contact with the deployment vessel (ie gauge wind direction and drift).

The unit is deployed by,

- 1) Set RF Channel, Depth, and Operating Mode
- 2) Remove and Discard Wind Flap and Parachute Sleeve
- 3) Remove and Open Parachute (Note, the parachute is used to slow the descent of the unit)
- 4) Toss unit over side so that the nose of buoy impacts water first. The toss also should help parachute to fill and open.

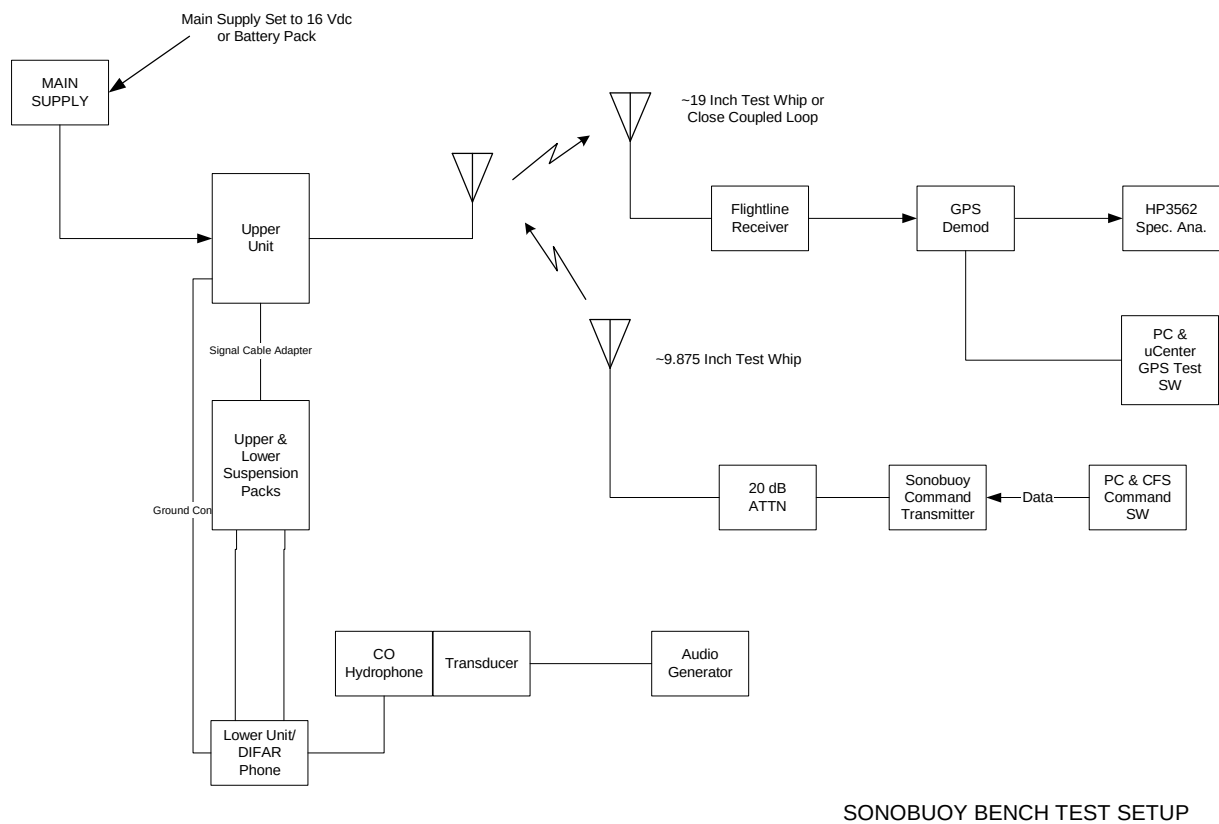
It is preferred that the deployment of the unit should be executed from the fan-tail of the deployment vessel while moving at a ~5-10 knots. The float will surface in about 18-20 seconds.

To recover the unit, grasp and raise the surface unit float and battery out of water. The recovery personnel should be careful not to make physical contact with sea-water ground when holding the external metal parts of the surface float (the current to lower is not very high, ~100 mA, but this can be a exciting experience). Disable battery power connection to surface unit by cutting the seawater battery ground wire (black wire). Pull the remaining hardware up by carefully pulling on the signal cable. Care should be exercised to keep vertical orientation of cable and pull tension below 20 lbs Note, the pull on this cable is resisted by deployed sea anchor that is located near the hydrophone. Additionally, the drift

of the recovery vessel should be controlled. It is recommended that the signal cable and bungee be wound onto a recovery aid to eliminate tangling (this easily occurs during recovery).

5.3 Standard Bench Operating Setup

The sonobuoy can be tested on the bench using the setup shown in Figure 7. To conduct the test, activate the unit using the selected main supply. The transmitter should be shut off using the CFS command generator approximately 60 seconds after power up. VHF reception can still be accomplished with the transmitter turned off by close coupling the buoy receive antenna and float. Some experimentation will be required to set this up the first time due to lab environment (distance should ~1-10 feet).



SONOBUOY BENCH TEST SETUP

Figure 7. GPS Sonobuoy Bench Test Setup

APPENDIX A
EXCERPTS FROM PSS
(PRODUCTION SONOBUOY SPECIFICATION)

**The following has been modified to incorporate selection of CO
bandwidths.**

3.3.5.1 Electronic function selector (EFS). Each buoy shall have an EFS for selecting the functions listed as per [Table XXII](#). The EFS shall be capable of performing at least 50 programming sequences and 100 verification sequences. Following launch, the EFS shall not be capable of being reprogrammed. The buoy shall operate in the modes selected by the EFS upon buoy activation unless overridden by an aircraft UHF command.

Table XXII. EFS Functions

Function	53F
VHF Channel	X
Operating Life	X
Select Sensor	X
AGC Mode	X
Specific Operating Depth	X
CO operating Band	X

b. Controls. Two push-button switches shall control the EFS. The left switch shall be the "SET" switch, and it shall enable function selection. The right switch shall be the "VERIFY" switch, and it shall cause the display of function selections stored in the EFS memory. The switch surface that must be pressed shall have a minimum diameter of 0.185 inches and no sharp edges. Switches shall provide tactile feedback when activated, if the entire surface of the switch is not the active area. Activation of the EFS switches through the SLC setting port(s) shall require no more than 8 pounds of force. The switch shall activate when pressed on each and every time it is pressed within the limits specified above.

d. Programming sequence. EFS programming shall be accomplished through use of the SET control. Each activation of the SET control shall select the displayed function's option and cycle the display to the next function's options. The programming sequence shall be in accordance with [Tables XXIV](#). Function selections shall start at the lowest or first value specified in [Table XXIV](#) and cycle to the next selection every 1 (+0.5, -0.1) sec. When selecting the VHF channel, the selected ten's digit shall remain displayed until the unit's digit has also been selected. The available VHF channels shall conform to [Tables XI](#) and [XII](#). Only upon the final SET control activation shall the EFS memory be updated with all the new selections except for the SSQ-53F, which shall update the settings of the memory for channel, life and depth (leaving the sensor and AGC mode unchanged), if the verify button is pressed after the depth selection is made (instead of selecting the sensor with the set button). The factory default settings for sensor and AGC mode shall be DIFAR and OFF. No selection of a function shall be possible until it is displayed. At least two cycles of values shall be displayed for each selection of a function, unless interrupted by a SET control activation. The programming sequence shall be enabled for a maximum of 60 sec.

e. Verification. Activation of the VERIFY control shall cause the function selections stored in the EFS memory to be displayed on the EFS Display. The stored settings shall be displayed only once for each activation of the VERIFY control. The time required to initiate the verification sequence shall not exceed 0.5 sec after the VERIFY control is activated.

2) SSQ-53F verification. The stored settings shall be displayed as per [Table XXIV](#) in the following sequence: VHF Channel, operating life, operating depth, operating sensor, and AGC mode. Displays shall cycle through the setting every 1.0, +0.5, -0.1 sec. The verification capability shall be disabled during the EFS programming sequence.

f. Memory. The EFS memory shall store selections for a minimum of 48 hours following activation of either the SET or VERIFY switches

Table XXIV EFS Sequence

Activation	53F	EFS Setting Options
1 st	channel (tens digit)	0-9
2 nd	channel (units digit)	0-9
3 rd	operating life	0.5, 1.0, 2.0, 4.0, 8.0
4 th	operating depth	d1, d2, d3, or d4
5 th	operating sensor	CS - CSO dF – DIFAR CO – Calibrated Omni
6 th	AGC mode	OF – Off A1 – Slow A2 – Fast
7 th	CO Bandwidth	o1 – 20 KHz BW o2 – 40 KHz BW
8 th	save to memory	

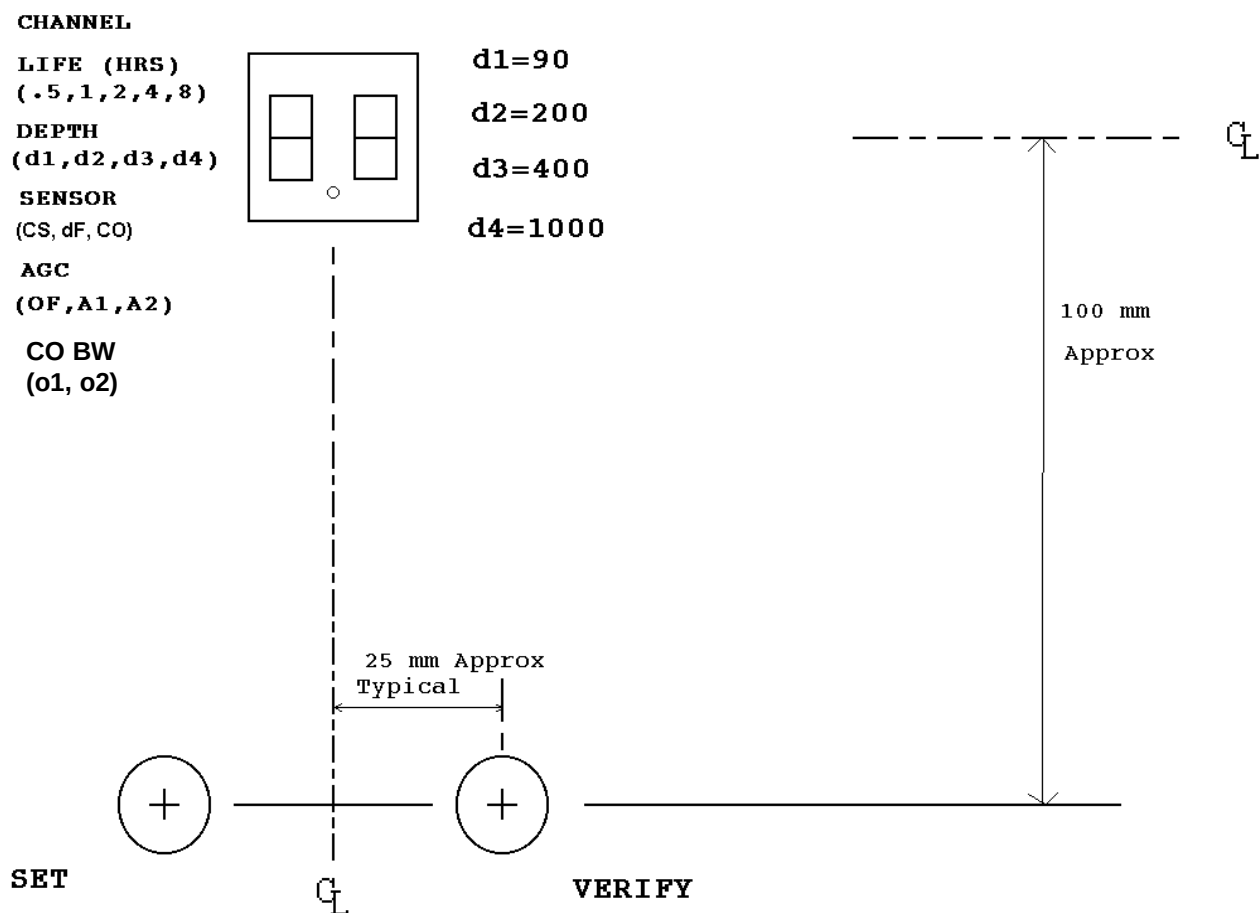


Figure 7B.

SSQ-53E EFS Displays and Controls

Type & Channel	Reference Input Frequency	RMS Sound Pressure Level above 1μPa	Peak Frequency Deviation Produced
53F/GPS Cal. Omni	100 Hz	116 ±2 dB	±19 kHz

Table XXX Acoustic Sensitivity

53F/GPS Cal. Omni	5-20000	40 kHz	5%
	5-20000	75 kHz	10%
	5-40000	40 kHz	5%
	5-40000	75 kHz	10%

Table XVI. Distortion

APPENDIX B - GPS UPLINK MESSAGES

GSA - GPS DOP and Active satellites.

GPS receiver operating mode, satellites used for navigation, and DOP values.

```
$GPGSA,A,3,23,29,07,08,09,18,26,28,,,,,1.94,1.18,1.54*0D
```

Name	ASCII String		Units	Description	
	Format	Example			
\$GPGSA	string	\$GPGSA		Message ID	GSA protocol header
Smode	character	A		Smode	See first table below
FS	1 digit	3		Fix Status	See second table below
Sv	numeric	23		Satellites Used	SV on Channel 1
Sv	numeric	29		Satellites Used	SV on Channel 2
...				...	Repeated for each channel
Sv	numeric			Satellites Used	SV on Channel 12
PDOP	numeric	1.94		PDOP	Position Dilution of Precision (00.0 to 99.9)
HDOP	numeric	1.18		HDOP	Horizontal Dilution of Precision (00.0 to 99.9)
VDOP	numeric	1.54		VDOP	Vertical Dilution of Precision (00.0 to 99.9)
cs	hexadecimal	*0D		Checksum	
				<CR> <LF>	End of message

Value	Description
M	Manual - forced to operate in 2D or 3D mode
A	allowed to automatically switch 2D/3D

Value	Description
1	Fix not available
2	2D Fix
3	3D Fix

See also [Position Fix Flags in NMEA Mode](#)

Note: If less than 12 SVs are used for navigation, the remaining fields are left empty. If more than 12 SVs are used for navigation, only the IDs of the first 12 are being output.

The SV Numbers (Fields 'Sv') are in the range of 1 to 32 for GPS satellites, and 33 to 64 for SBAS satellites (33 = SBAS PRN 120, 34 = SBAS PRN 121 and so on)

Example:

```
$GPGSA,A,3,23,29,07,08,09,18,26,28,,,,,1.94,1.18,1.54*0D
```

GSV - GPS Satellites in View

The number of satellites in view, together with each PRN (SV Id), elevation and azimuth, and C/No (Signal/Noise Ratio) value. Only four satellite details are transmitted in one message. There are up to 4 messages used as indicated in the first field NoMsg.

```
$GPGSV,NoMsg,MsgNo,NoSv,{,sv,elv,az,cno}*cs<CR><LF>
```

Name	ASCII String		Units	Description	
	Format	Example			
\$GPGSV	string	\$GPGSV		Message ID	GSV protocol header
NoMsg	1 digit	3		Number of Messages	Total number of GPGSV messages being output
MsgNo	1 digit	1		Message Number	Number of this message
NoSv	numeric	10		Satellites in View	
sv	numeric	23		Satellite ID	SV ID (GPS: 1-32, SBAS 33-64 (33=PRN120))
elv	numeric	38	degree	Elevation	Maximum 90
az	numeric	230	degree	Azimuth	Range 0 to 359
cno	numeric	44	dBHz	C/No	Range 0 to 99, null when not tracking
...				...	Data of 2nd, 3rd Satellite (same as above)
sv	numeric	29		Satellite ID	SV ID (Range 1 to 32)
elv	numeric	71	degree	Elevation	Maximum 90
az	numeric	156	degree	Azimuth	Range 0 to 359
cno	numeric	47	dBHz	C/No	Range 0 to 99, null when not tracking
...				...	
cs	hexadecimal	*7F		Checksum	
				<CR> <LF>	End of message

Example:

```
$GPGSV,3,1,10,23,38,230,44,29,71,156,47,07,29,116,41,08,09,081,36*7F
$GPGSV,3,2,10,10,07,189,,05,05,220,,09,34,274,42,18,25,309,44*72
$GPGSV,3,3,10,26,82,187,47,28,43,056,46*77
```

GGA - GPS fix data

Time and position, together with GPS fixing related data (number of satellites in use, and the resulting HDOP, age of differential data if in use, etc.).

```
$GPGGA,hhmmss.ss,Latitude,N,Longitude,E,FS,NoSV,HDOP,msl,m,Altref,m,DiffAge,DiffStation*cs<
```

Name	ASCII String		Units	Description	
	Format	Example			
\$GPGGA	string	\$GPGGA		Message ID	GGA protocol header
hhmmss.ss	hhmmss.sss	092725.00		UTC Time	Current time
Latitude	dddmm.mmmm	4717.11399		Latitude	Degrees + minutes
N	character	N		N/S Indicator	N=north or S=south
Longitude	dddmm.mmmm	00833.91590		Longitude	Degrees + minutes
E	character	E		E/W indicator	E=east or W=west
FS	1 digit	1		Position Fix Indicator	See Table below
NoSV	numeric	8		Satellites Used	Range 0 to 12
HDOP	numeric	1.01		HDOP	Horizontal Dilution of Precision
msl	numeric	499.6	m	MSL Altitude	
m	character	M		Units	Meters (fixed field)
Altref	blank	48.0	m	Geoid Separation	
m	blank	M		Units	Meters (fixed field)
DiffAge	numeric		s	Age of Differential Corrections	Blank (Null) fields when DGPS is not used
DiffStation	numeric	0		Diff. Reference Station ID	
cs	hexadecimal	*5B		Checksum	
<CR> <LF>					End of message

Fix Status	Description
0	No fix / Invalid
1	Standard GPS (2D/3D)
2	Differential GPS
6	Estimated (DR) Fix

See also [Position Fix Flags in NMEA Mode](#)

Example:

```
$GPGGA,092725.00,4717.11399,N,00833.91590,E,1,8,1.01,499.6,M,48.0,M,,0*5B
```

Note: The output of this message is dependent on the currently selected datum (Default: WGS84)

Remarks:

According to the NMEA Standard, Latitude and Longitude are output in the format Degrees, Minutes and (Decimal) Fractions of Minutes. To convert to Degrees and Fractions of Degrees, or Degrees, Minutes Seconds and Fractions of seconds, the 'Minutes' and 'Fractional Minutes' parts need to be converted. In other words: If the GPS Receiver reports a Latitude of 4717.112671 North and Longitude of 00833.914843 East, this is

Latitude 47 Degrees, 17.112671 Minutes
Longitude 8 Degrees, 33.914843 Minutes

or

Latitude 47 Degrees, 17 Minutes 6.76026 Seconds
Longitude 8 Degrees, 33 Minutes 54.89058 Seconds

or

Latitude 47.28521118 Degrees
Longitude 8.56524738 Degrees

RMC - Recommended Minimum data

The Recommended Minimum sentence defined by NMEA for GPS/Transit system data.

```
$GPRMC, hhmmss, status, latitude, N, longitude, E, spd, cog, ddmmyy, mv, mvE, mode *cs <CR> <LF>
```

Name	ASCII String		Units	Description	
	Format	Example			
\$GPRMC	string	\$GPRMC		Message ID	RMC protocol header
hhmmss	hhmmss.sss	083559.00		UTC Time	Time of position fix
status	character	A		Status	Status V = Navigation receiver warning A = Data valid. See Position Fix Flags in NMEA Mode
latitude	ddmm.mmmm	4717.11437		Latitude	User datum latitude degrees, minutes, decimal minutes format
N		N		N/S Indicator	N=north or S=south
longitude	ddmm.mmmm	00833.91522		Longitude	User datum latitude degrees, minutes, decimal minutes format
E	character	E		E/W indicator	E=east or W=west
Spd	numeric	0.004	knots	Speed	Speed Over Ground
cog	numeric	77.52	degrees	COG	Course Over Ground
ddmmyy	ddmmyy	091202		Date	Current Date in Day, Month Year format
mv	numeric		degrees	Magnetic variation value	Not being output by receiver
mvE	character			Magnetic variation E/W indicator	Not being output by receiver
mode				Mode Indicator	See Position Fix Flags in NMEA Mode
cs	hexadecimal	*53		Checksum	
<CR> <LF>					End of message

Example:

```
Example: $GPRMC,083559.00,A,4717.11437,N,00833.91522,E,0.004,77.52,091202,,,A*57
```

Position Fix Flags in NMEA Mode

The following list shows the implementation of the ANTARIS NMEA protocol and under what circumstances which flag is set.

NMEA Message	Field	No Position Fix (after power-up, after losing Satellite lock)	Valid Position Fix, but User Limits exceeded	EKF (only on Dead Reckoning Receivers)	2D Position Fix	3D Position Fix	Combined GPS/EKF Position Fix (only on Dead Reckoning Receivers)
GGA	Status	0	0	6	1/2	1/2	1/2
		0=Fix not available/invalid, 1=GPS SPS Mode, Fix valid, 2=Differential GPS, SPS Mode, Fix Valid, 6=Estimated/Dead Reckoning					
GLL	Status	V	V	A	A	A	A
		A=Data VALID, V=Data Invalid					
GSA	Mode Indicator	N	N	E	A/D	A/D	A/D
		N=No Fix, A=Autonomous GNSS Fix, D=Differential GNSS Fix, E=Estimated/Dead Reckoning Fix					
GSA	Nav Mode	1	1	E	2	3	3
		1=Fix Not available, 2=2D Fix, 3=3D Fix, E=Estimated/Dead Reckoning Fix					
RMC	Status	V	V	A	A	A	A
		A=Data VALID, V=Data Invalid					
RMC	Mode Indicator	N	N	E	A/D	A/D	A/D
		N=No Fix, A=Autonomous GNSS Fix, D=Differential GNSS Fix, E=Estimated/Dead Reckoning Fix					
VTG	Mode Indicator	N	N	E	A/D	A/D	A/D
		N=No Fix, A=Autonomous GNSS Fix, D=Differential GNSS Fix, E=Estimated/Dead Reckoning Fix					

Note: Please note, that the NMEA standard does not fully conform with the above table, because NMEA is not prepared for combined GPS/Dead Reckoning Receivers. However, the allocation has been chosen such that inconsistencies are kept to a minimum.

It important to note that - by default - the receiver will not output invalid data. In that case, it will output empty fields. This is new with firmware version 3.0.

- A valid position fix is reported as follows:

```
$GPGLL,4717.11634,N,00833.91297,E,124923.00,A,A*6E
```

- An invalid position fix (but time valid) is reported as follows:

```
$GPGLL,,,,,124924.00,V,N*42
```

- If Time is unknown (e.g. during a cold-start),

```
$GPGLL,,,,,V,N*64
```

Note: In firmware versions older than 3.0, the receiver did output invalid data and marked invalid data with the 'Invalid/Valid' Flags. If needed, this behaviour can still be enabled on firmware version 3.0 and up, with the help of the UBX protocol message [CFG-NMEA \(0x06 0x17\)](#)

APPENDIX C - CFS SONOBUOY COMMANDS

AN/SSQ-53F GPS SONOBUOY VALID CFS COMMANDS/SUBCOMMANDS/DATA WORDS

Command Name	Command Word 1	Sub-Command Word 1	Data Word 11	Data Word 12	Data Word 21	Data Word 22
RF Power ON/OFF	17	X	0 "VHF Power Off" 4 "1 Watt"	0 Not Used	-	-
RF Channel	21	X	1-99 RF Channels	0 Not Used	-	-
Acoustic Gain/AGC	34	X	0 Not Used	0 AGC Off 1 Slow AGC On 2 Fast AGC On	-	-
Modulation Input	40	X	0 Not Used	0 DIFAR 1 CSO 2 Cal'd Omni	-	-
Life (Days)/ Life (1/4 Hrs)	68	X	0 0 Days	2 0.5 Hr 4 1.0 Hr 8 2.0 Hr 16 4.0 Hr 32 8.0 Hr	-	-
CO Bandwidth Select	3	35	111	0 Standard (20K) 1 Extended (40K)	-	-
GPS Subcarrier On/Off	3	35	110	0 Subcarrier Off 1 Subcarrier On	-	-
Lower Unit Power ON/OFF	3	35	8	0 - Power Lower Off 1 - Power Lower On	-	-

Note: Scuttle ID must match Fixed ID for valid Scuttle execution.